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RESEARCH ARTICLE

Measurements of turbulent premixed flame dynamics using cinema stereoscopic PIV

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Abstract A new experimental method is described that provides high-speed movies of turbulent premixed flame wrinkling dynamics and the associated vorticity fields. This method employs cinema stereoscopic particle image velocimetry and has been applied to a turbulent slot Bunsen flame. Three-component velocity fields were measured with high temporal and spatial resolutions of 0.9 ms and 140 μm, respectively. The flame-front location was determined using a new multi-step method based on particle image gradients, which is described. Comparisons are made between flame fronts found with this method and simultaneous CH-PLIF images. These show that the flame contour determined corresponds well to the true location of maximum gas density gradient. Time histories of typical eddy–flame interactions are reported and several important phenomena identified. Outwardly rotating eddy pairs wrinkle the flame and are attenuated at they pass through the flamelet. Significant flame-generated vorticity is produced downstream of the wrinkled tip. Similar wrinkles are caused by larger groups of outwardly rotating eddies. Inwardly rotating pairs cause significant convex wrinkles that grow as the flame propagates. These wrinkles encounter other eddies that alter their behavior. The effects of the hydrodynamic and diffusive instabilities are observed and found to be significant contributors to the formation and propagation of wrinkles.

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1 Introduction

Turbulent premixed combustion has gained renewed interest recently due to its importance in the development of low emission engines for transportation and power generation (Cabot et al. 2004; Hayashi et al. 2005; Johnson et al. 2005). However, there are many unanswered questions regarding the fundamental dynamics of such flames. It is generally believed that for a wide range of conditions the flame front resembles an unsteady, stretched, laminar flame. In this flamelet regime, the flame consists of thin sheets that are wrinkled by the turbulence. That is, the instantaneous topography of these sheets is determined by the detailed history of the turbulent eddies that interact with them. Important properties such as the fuel consumption rate, statistical flame shape and position, and the local chemical reaction details are determined by this topography and the related unsteady strain rate on the flame surface. Hence, in order to properly understand and model turbulent flamelets, we must understand the dynamics of the unsteady turbulence–flame interaction.

The fundamental unit of the turbulence–flame interaction is the eddy–flame interaction. If we consider the flame to be a wave propagating at a fixed speed, the simplest wrinkling mechanism is apparent: velocity gradients in the reactants will cause certain flame segments to travel faster or slower in the laboratory flame. In a quasi-steady case, the flame will position itself such that the reactant velocity normal to the surface equals the propagation speed,

resulting in wrinkles. Damköhler (1940) originally postulated that the velocity gradients implied by a simple theoretical vortex would produce a pair of conical wrinkles. The increase in fuel consumption rate between turbulent and laminar flames was predicted to be due to the increase in reaction surface area. Since then, this concept has been modified to reflect more practical scenarios by including phenomena such as inefficiencies in the stretching process (Meneveau and Poinsot 1991; Colin et al. 2000; Charlette et al. 2002), the effects of strain and curvature (Chen and Im 2000; Hawkes and Cant 2000; Chakraborty and Cant 2005), and instabilities arising from hydrodynamic or diffusive effects (Kobayashi and Kawazoe 2000; Yuan et al. 2006).

In order to understand these phenomena and the modifications they make to the wrinkling process, canonical studies have been performed involving the interaction between simple 2D vortex structures and a laminar flame front. Common configurations for premixed flame experiments include planar wake generated vortices impinging on a V-flame (Lee et al. 1993, 1995; Nye et al. 1996; Hargrave and Jarvis 2006), and interactions between a 2D outwardly rotating vortex pair and a laminar flat flame (Roberts and Driscoll 1991; Mueller et al. 1998; Sinibaldi et al. 1998) or counterflow flame (Renard et al. 1998). These experimental studies have quantified the effects of unsteady strain and curvature on both the topography and chemistry of the flame. Furthermore, the attenuation and generation of vorticity by the flame front has been investigated. The intrinsic instabilities of laminar premixed flames have also been studied by many authors (e.g. Sivashinsky 1977; Denet and Haldenwang 1992; Kadowaki 1999).

While these studies provide information about fundamental processes, studies of fully turbulent flames are needed to understand the more complex case of turbulent flame dynamics. In real turbulence the isolated vortex pair–laminar flame interaction does not occur. Instead, a chaotic sequence of turbulent eddies impinge on a dynamically curved and strained flame front. Long chains of eddies may create large scale wrinkles or may have counteracting effects and cause negligible wrinkling. Inwardly rotating eddy pairs may stretch the flame in unpredicted ways. Parameters such as the interaction angle and inter-eddy spacing may also play an important role. Furthermore, the 2D isolated vortex configuration does not allow for many of the effects controlling the evolution of 3D turbulence; there is no inter-scale energy transfer or vortex line stretching. Without these effects, the behavior of the vortices, particularly at small scales, will not be accurately represented. Hence, in order to understand the ability of real turbulence to wrinkle premixed flames, we must study their interaction directly.

Recently, Filatyev et al. (2005) obtained instantaneous images of eddy–flame interaction in a turbulent slot Bunsen flame using simultaneous PIV and CH-PLIF diagnostics. They showed what may be isolated cases of wrinkling phenomena that are similar to those found in the laminar studies. Filatyev et al. (2007) went further, using double pulsed stereoscopic PIV to determine statistical fluxes of the mean progress variable. Acetone PLIF was used to locate the flame contour. Tanahashi et al. (2005) combined simultaneous CH-OH PLIF with stereoscopic PIV in a turbulent premixed swirl-stabilized flame. They demonstrated the need for three-component velocity measurements in order to interpret flame wrinkling phenomena.

However, these studies only recorded isolated or double shot images; they did not obtain the time-resolved information needed to understand the unsteady dynamics of a turbulent flame in detail. Wrinkling has been observed at scales ranging from the turbulence micro-scales (Tanahashi et al. 2005) to that of the full burner (Filatyev et al. 2005). Hence, an understanding of the mechanisms causing flame wrinkling over this entire range is needed. That is, we must observe the causality between the turbulence field and the flame front dynamics with both high temporal and spatial resolution. Furthermore, all velocity components should be measured in order to avoid erroneous interpretations. These velocities should be measured simultaneously in the reactants and products, allowing the full interaction history to be recorded. These objectives have not been accomplished by previous studies.

To achieve them, a new cinema stereoscopic particle image velocimetry (CS-PIV) system, capable of recording three-component velocity fields at several kilohertz with a spatial resolution of 140 μm (interrogation boxes of 280 μm) has been developed. This system has been applied to a 2D methane–air slot Bunsen flame and is described below. A predictor–corrector method of mapping the flame surface topography from the gradient of particle image density is presented and its accuracy analyzed using simultaneous Mie scattering/CH-PLIF images. The global flame and flow statistics are reported and the effects of diffusive stability shown to be significant. Typical eddy/flame interaction events are presented and important phenomena discussed. Outwardly rotating eddy pairs are observed to cause wrinkles concave towards the reactants that burn through into cusps. The eddies are attenuated by the flame and flame-generated vorticity (FGV) is created in the products. Similar wrinkles are formed by larger groups of eddies. Inwardly rotating eddy groups are found to create convex wrinkles that grow, encountering further eddies that affect their behavior. The hydrodynamic and preferential diffusion effects on the flame stability are observed and found to be significant in the formation and propagation of wrinkles.

2 Cinema stereoscopic PIV

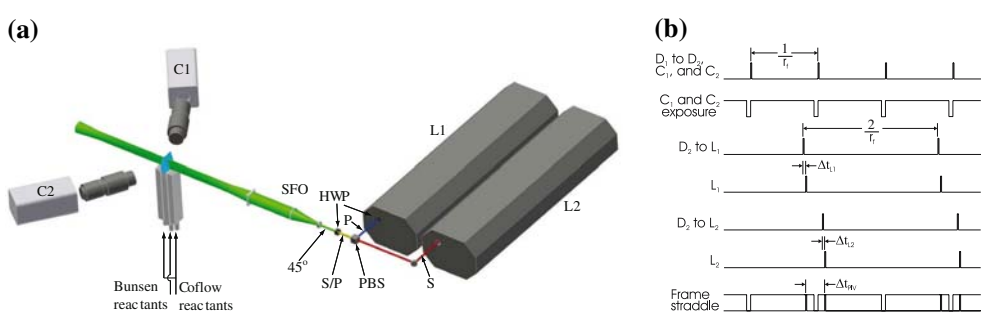
Development of the cinema stereoscopic PIV system involved simultaneous implementation of various conventional PIV techniques. Cinema PIV measurements in turbulent non-premixed flames have been reported by Upatnieks et al. (2004). Cinema-stereoscopic PIV has been used by Ganapathisubramani et al. (2007) with the laser plane normal to the flow direction for the study of non-reacting turbulence. Employing CS-PIV for the detailed study of flame dynamics demanded a balance between conflicting requirements while achieving sufficient accuracy all three velocity components. That is, we required a field of view and frame rate sufficiently large to track the progress of eddies and capture the resulting flame motion, but sufficient resolution to identify the smallest scale turbulent structures. The signal quality was dependant on the particle image quality, and hence on the focus and light collection ability of the camera/lens system. This was greatly effected by the low pulse energy of high-rep-rate lasers and the requirement to filter chemiluminescence from the flame. Hence, it was necessary to optimize several factors including camera resolution, camera and laser frame rate, magnification, flow speed, stereo configuration, sheet dimensions, and PIV particle properties.

A schematic of the CS-PIV system is shown in Fig. 1a. An angular arrangement satisfying the Scheimpflug criterion was employed. The angle between the laser sheet normal and the lens axis was 35°. The cameras were placed in a dual forward scatter configuration to increase light gathering and maximize the useful field of view. The two cameras used were Phantom v9.0 s (Vision Research), which posses a maximum resolution of 1,632 × 1,200 11.5 μm square pixels at frame rates of over 1 kHz. Higher frame rates are possible at lower resolution. A pair of ORC-1000 high-rep-rate Nd:YAG lasers (Clark-MXR), capable of producing 8 mJ per pulse of 532 nm light at several kilohertz, were used to create the light sheets. The beams from the two lasers were polarized orthogonally to each other and combined using a polarization based system. This avoided the high losses common to non-polarizing systems. To eliminate the difference in scattering from the S- and P-beams, the polarization was then rotated 45° by a half wave plate. A sufficiently thin sheet was produced by passing the laser beam through a 5× Galilean telescope. This provided sufficient vertical and horizontal expansion to create a sheet with a full-width at half-maximum of approximately 300 μm and a usable height of 20 mm.

A balance between field of view, frame rate, resolution, and flow speed was found with the cameras operating at a resolution of 672 × 1,120 pixels and frame rate (r_f) of 2,222 Hz. Particle image pairs of a 12.8 mm × 18.2 mm field of view were obtained every 0.9 ms. With the long axis aligned with the principal flow direction, this allowed approximately 13 images of an eddy convecting at 1.5 m/s to be obtained. Frame straddling was employed and the time delay (Δt_{PIV}) between correlated particle images was set by the laser pulses as 40 μs. A timing diagram describing the operation of the system is provided in Fig. 1. Data was collected in spans of approximately 1 s, resulting in over 4,000 images and 3.5 GB of data per run.

The flow was seeded with silica coated titanium dioxide particles (Huntsman A-PP2). These particles had a nominal diameter of 0.15 μm and survived passage through the flame. It was found that these particles possessed a much narrower range of particle diameters than other solid, high temperature flow tracers (e.g. alumina) due to less agglomeration. The seed density was adjusted in real-time to provide an optimal balance between valid PIV vectors in the reactants and products. That is, the seed density was set to be slightly higher than optimal in the reactants, resulting in a slightly lower than optimal level in the products after expansion. This scheme, combined with an adaptive multi-pass interrogation method, provided sufficient particles for accurate vector determination throughout the flow field. Scattered light from the particles was collected by the cameras using Micro-Nikkor 105 mm lenses operating at f/8.0. Narrow bandpass filters were placed in front of the lenses to remove chemiluminescence from the flame. A

Fig. 1 CS-PIV system



System layout. Polarization indicated. HWP - Half wave plate, PBS - Polarizing beam splitter, SFO - Sheet forming optics.

Timing diagram. D_i are delay generator signals.

summary of the optimized CS-PIV parameters is given in Table 1.

The angular configuration of the CS-PIV system created a strong perspective distortion. That is, in order to maintain focus across the laser sheet at an acute viewing angle, the object, lens, and camera planes had to meet at a single point as described by the Scheimpflug criterion (Raffel et al. 1998). The differences in orientation between these planes resulted in varying distances between the laser sheet and camera sensor. This resulted in non-uniform magnification across the field of view. To calibrate the cameras for this, a target was placed in the light sheet and imaged at various z-locations. A specially designed target was required, as standard targets were found to cause unacceptable errors due to two system attributes. The dual-forward scatter configuration required simultaneous imaging from opposite sides of the target. Additionally, the high resolution of this system required high magnification collection optics, resulting in a limited depth of field. Standard two-sided targets created noticeable translation and focusing errors due to their thickness. Transparent targets marked on a single side caused similar errors due to light bending effects in the target medium. To avoid these errors, a thin, translucent, symmetric calibration target was used. This was comprised of a thin film, marked with 300 μm crosses on a 750 μm grid and sandwiched between 100 μm thick glass flats. This created equal in-plane shifts for both cameras and out-of-plane shifts an order of magnitude smaller than the depth of field.

The distorted images from the cameras were corrected to a normal coordinate system and registered using the pin-hole camera model in the commercial DaVis software package by LaVision. This process mapped equivalent pixels from each camera to the same physical position in the measurement plane. Minute adjustments were made to ensure proper registration of the cameras using the DaVis stereoscopic self calibration method, which ensures that simultaneous particle fields from both cameras contain identical information. The spatial cross correlation and consequent velocity components were computed using the same software. A multi-pass vector evaluation technique was employed with interrogation box sizes decreasing from 32 × 32 to 16 × 16 pixels with a 50% overlap. This resulted in a roughly cubical interrogation volume with a box size of approximately 280 μm and a vector spacing of approximately 140 μm. The adequacy of the dynamic range to accurately measure both reactant and product velocities was confirmed by systematically varying Δt_{PIV}. It was found that the conditional statistics were unaltered by this parameter in the range of the selected value. The number of spurious vectors was less than 1.5% in both the reactants and products.

Table 1 CS-PIV parameters

Parameter	Value
Camera frame rate	2,222 Hz
Temporal resolution	0.9 ms
Δt_{piv}	40 μs
Stereo angle	35°
Camera resolution	672 px × 1,120 px
Field of view	12.8 mm × 18.2 mm
Interrogation box	280 μm
Vector spacing	140 μm
Seed material	Silica coated TiO_2
Nominal seed diameter	0.15 μm

Laser Doppler Velocimetry measurements were made to determine the appropriate spatial resolution requirement. Using the Taylor scale (λ_{T}) as the desired resolution, it was determined that accurate tracking of eddies with a characteristic size of 0.5 mm was required. The temporal Taylor scale was determined by fitting an osculating parabola to the velocity autocorrelation:

$$\lambda_{\mathrm{T}}^2 = \frac{-2}{\left(\frac{d^2\Re}{dt^2}\right)_{t=0}} \tag{1}$$

where $\Re$ is the autocorrelation given by:

$$\Re(t) = \frac{\overline{u''(\underline{x})u''(\underline{x}+t)}}{\overline{u''(\underline{x})^2}} \tag{2}$$

Here, u'' is the local instantaneous velocity fluctuation. The equivalent length scale was determined using Taylor's hypothesis.

The minimum eddy size properly resolved by the CS-PIV was determined by tracking the small structures in time. The eddy size was determined by fitting an osculating parabola to the local velocity fluctuation correlation for the particular eddy. If an eddy was present for a minimum of two frames in a region of low out-of-plane velocity, it was considered to be properly resolved. It was determined that eddies with a characteristic size of 0.5 mm were computed accurately in approximately 75% of the cases. The remaining eddies were either dissipated, passed out of the interrogation plane between frames, or were noise.

3 Experimental configuration

The flame studied was stabilized on the collection of three slot burners shown in Fig. 2. The Bunsen flame of interest was stabilized on the central burner, which provided the flame with methane–air mixtures of various equivalence ratio at a bulk velocity of 1 m/s. Two outer burners

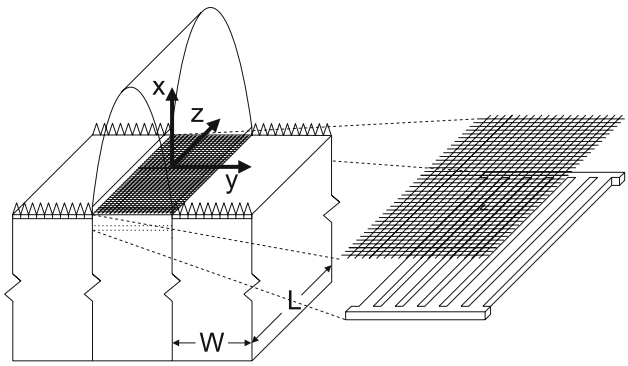


Fig. 2 2D slot bunsen burner with side burners. $W = 25$ mm, $L = 50$ mm. The origin of the coordinate system is centered in the middle burner at the x-plane in which the flame is anchored

Table 2 Test conditions

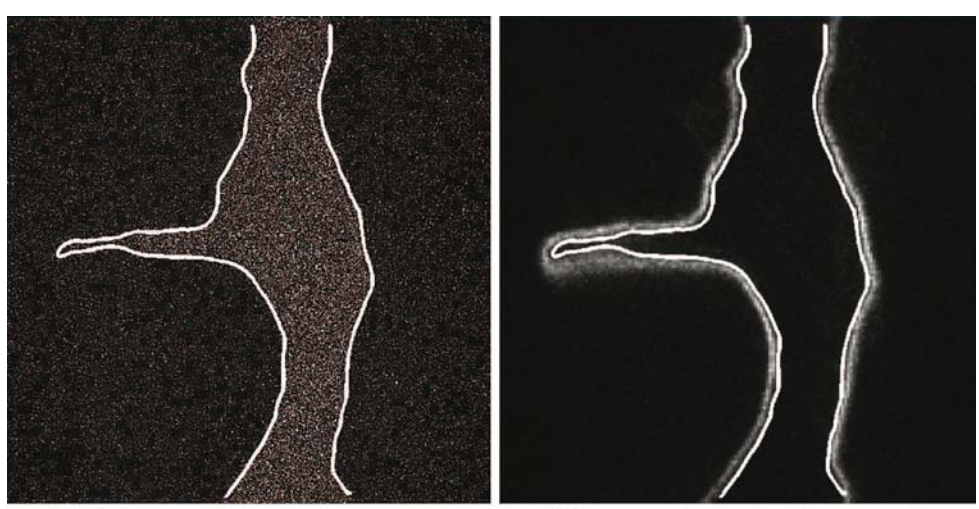
Case	ϕ	u'/S_{L0}	Λ/l_f	Ma	Le
1	0.6	3.8	6.9	<0	1.0
2	0.7	2.3	11.5	≈0	1.0
3	1.35	2.3	11.5	>0	1.0

produced short, grid stabilized flames. The flow rate through the side burners was set such that the mean downstream velocity profiles of the products in both the central and side burners closely matched. This minimized the effects of shear on the flame dynamics. Flow rates were controlled using metered choked orifices and the experiment was conducted at approximately 295 K and 1 atm.

The primary turbulent fluctuations in the central burner were generated by means of a slot grating placed 1 cm upstream of the flame stabilization plane. A loose wire mesh was installed downstream of this grating to break up the large recirculation zones and prevent flame stabilization on the grating. This combination provided strong fluctuations in all directions, with no mean out-of-plane velocity. Details of the flow field are given in Sect. 6.

The equivalence ratio was varied between 0.6 and 1.35. Specific conditions are given in Table 2. The conditions were selected to highlight the different regimes of preferential diffusive instability associated with lean and rich methane–air flames and characterized by the Markstein number (Ma). As a flame is stretched, differential diffusion between the different reactants and/or heat causes certain flame segments to accelerate or decelerate. The correlation between the stretch rate and this distribution of flame speeds causes flame wrinkles to either grow or shrink. A negative (positive) value of Ma is associated with a diffusively unstable (stable) flame. The specific equivalence ratio for Case 3 was selected such that the unstretched laminar flame speed (S_{L0}), as computed by Chemkin, was the same as Case 2. The methane–air mixtures tested possessed Lewis numbers of approximately unity. This effectively eliminated the thermo-diffusive effects of stretch rate on the flame speed. Hence, variations in the flame speed were related to preferential-diffusive as opposed to thermo-diffusive imbalances. Furthermore, Case 2 ($Ma \approx 0$) was selected such that the local flame speed remained essentially unchanged regardless of the stretch rate.

The regime of turbulent premixed combustion in which these flames fall can be characterized by the laminar flame speed, flame thickness, turbulence velocity scale, and turbulence length scale in various combinations (Peters 2000). Here, we use the turbulence intensity, defined as the ratio between root-mean-squared velocity fluctuations in the downstream direction (u') and the laminar flame speed, and



(a) Maximum particle gradient contour **(b)** Correspondence with CH-layer location

Fig. 3 Results from particle gradient based flame finding method

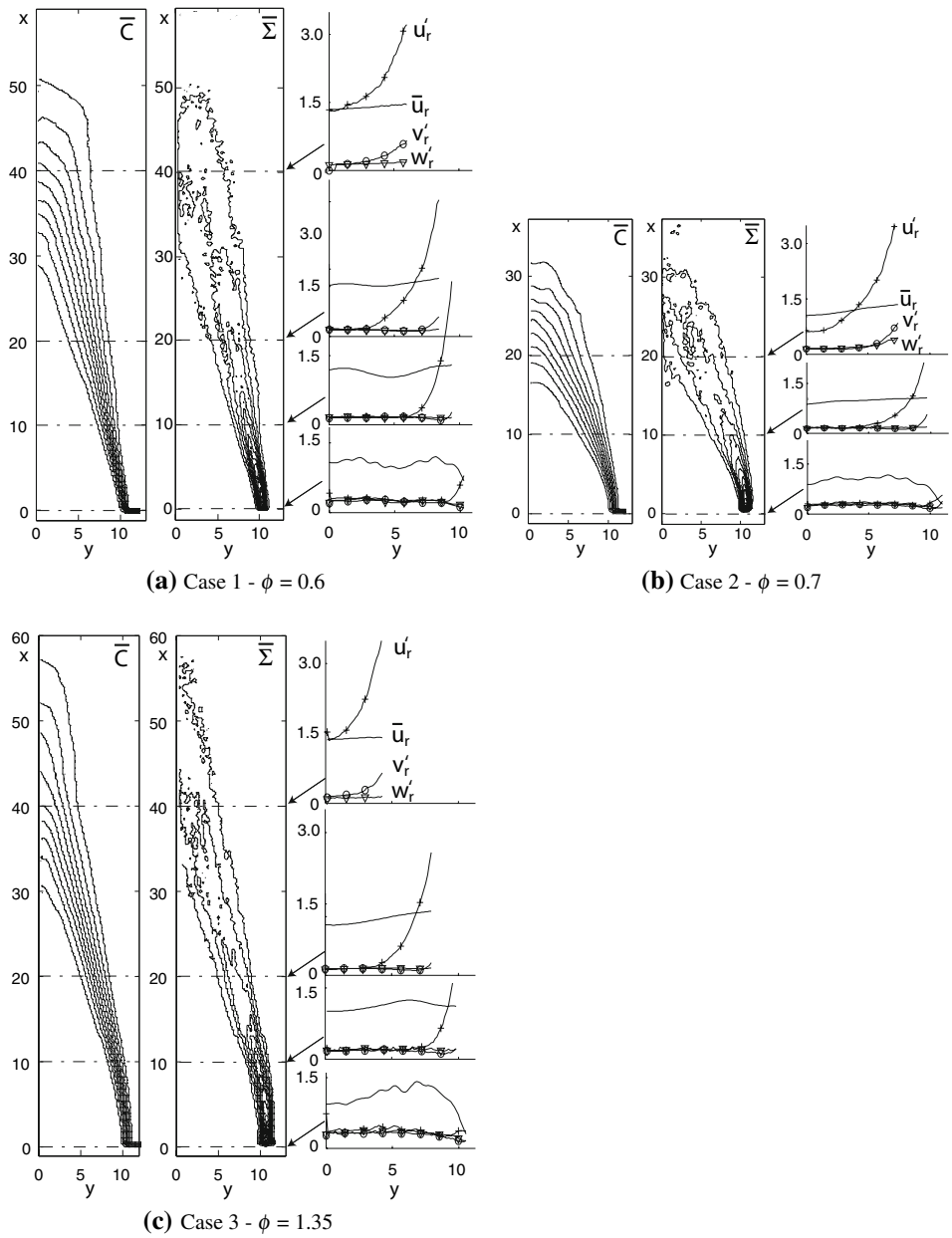


Fig. 4 Mean reactedness ($\overline{C}$) contours are from 0.1–0.9 in 0.1 increments. Flame surface density ($\overline{\Sigma}$) contours range from 0.1 mm^{-1} to 1.2 mm^{-1} in 0.1 mm^{-1} incriments. Velocity statistics are conditioned on the reactants. $(-) - \overline{u_r}, (+) - u_r', (\circ) - v_r', (\nabla) - w_r'$. Velocities are in m/s. Positions are in mm

the ratio between the longitudinal turbulence integral scale (Λ) and the flame diffusive thickness defined as:

$$l_f = \frac{\alpha_u}{S_{L0}} \tag{3}$$

where α_u is the thermal diffusivity of the reactant mixture at the unburnt temperature. This thickness approximately represents the width of the reaction zone. The integral scale was computed from the integral of the autocorrelation in Eq. 2. The turbulence statistics are reported at the burner exit. However, since the turbulence changes as it progresses downstream, this does not completely characterize the turbulence where it interacts with the flame. The details of the flow field are presented in Sect. 6.

4 Flame front mapping

In order to observe the interaction between eddies and the flame, the instantaneous flame front position had to be determined. Simultaneous measurements of the velocity field and premixed flame properties have been performed by combining PIV with PLIF (Filatyev et al. 2005; Tanahashi et al. 2005; Cho et al. 2004) and Rayleigh scattering (Pfadler et al. 2005) techniques. However the use of these methods is not currently feasible in high-rep-rate applications due to laser limitations. Therefore, a method to determine the flame front location from the particle images was developed and its accuracy evaluated.

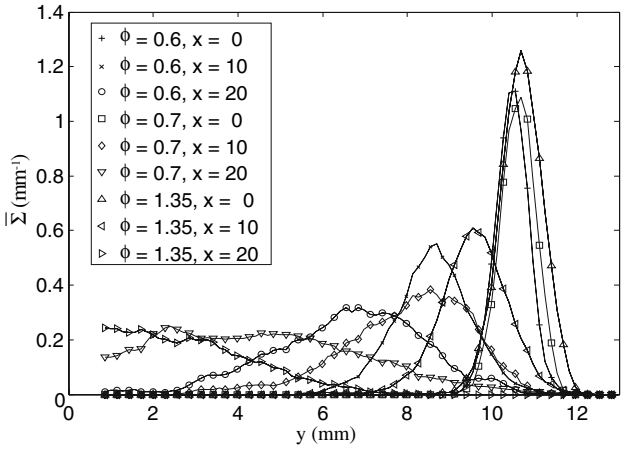


Fig. 5 Profiles of flame surface density

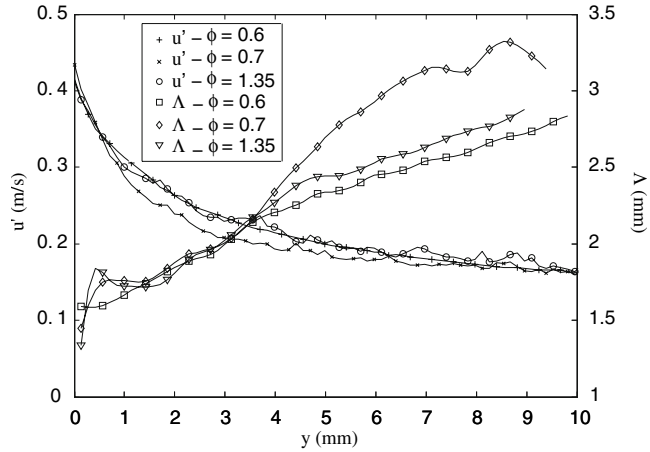


Fig. 6 Centerline turbulence statistics

Flame front location was accomplished by measuring the dilatation of the gas from the gradient of seed particle number density. Typical methods of mapping a flame front from seed density have generally involved threshold techniques (Stella et al. 2001; Pfadler et al. 2005). However, these techniques rely on an ad hoc threshold value and are constrained by the spatial filter selected, which is typically isotropic. These problems are greatly compounded when using solid seed; the particles survive the flame, creating a gradient of number density instead of a sharp cutoff. However, this gradient itself can be used to accurately identify the flame surface. That is, we can associate the flame surface with an iso-contour of particle image density gradient. Here we use the maximum gradient contour. However, the highly non-uniform nature of the particle field made accurate determination of these gradients from a raw image very difficult; isolated seed density variations and high-frequency noise appeared as false expansion contours in the reactants. Therefore, a two-step method was used. First, an initial predictor step with a standard filter scheme was applied along with an adaptive threshold technique. The second step repositioned the flame to coincide with the center of the local high gradient region using an edge finding filter aligned normal to the flame boundary predicted in the first step. As the density gradient curve had an S-shaped profile, this corresponded to the location of the maximum gradient. This two-step procedure effectively eliminated noise due to variations in particle seed density as shown in Fig. 3.

In order to quantify the accuracy of this method, comparisons were made between flame fronts found from the particle density gradient with those corresponding to the center of the CH-layer. This was done using simultaneous Mie scattering/CH-PLIF images originally taken by Filatyev et al. (2005). As can be seen from Fig. 3, the particle gradient contour is slightly offset from the CH-layer towards the reactants. The mean offset from the center of the CH-layer was 435 μm and the standard deviation was 50 μm. The relatively high standard deviation was caused by local thickening of the CH layer due to out-of-plane curvature. In regions where the CH layer thickness was within 25% of the mean, the mean offset was 425 μm with a standard deviation of 25 μm.

This difference between the location of the particle image flame front and the CH-layer is expected. The particle density is proportional to the gas density. Therefore, the center of the high particle density gradient region should correspond to that of the high gas density gradient region. Chemkin simulations of a laminar methane–air flame indicate that the maximum density gradient should occur approximately 415 μm upstream of the center of the CH-layer. The simulations also indicate that the high density gradient region occurs slightly ahead of the high temperature and velocity gradient regions due to dissociation. This, combined with the low Stokes number of the sub-micron seed, resulted in an expected accumulated particle lag due to thermophoretic forces (Sung et al. 1994) and acceleration of less than 20 μm. Thus it was concluded that the location of the measured maximum particle gradient contour corresponds to the maximum gas density gradient contour with an uncertainty of less than 25 μm.

5 Results

5.1 Flame characterization

Shown in Figs. 4, 5 are global statistics of the flame and flow field taken from 1,000 images. The mean progress variable ($\overline{C}$) was calculated based on an infinitely thin flame assumption. That is, $C(t)$ was unity in the products and zero in the reactants. The interface was defined from

the density gradient contour described above. The flame surface density ($\overline{\Sigma}$) was defined as the mean flame surface area per unit volume. For a 2D measurement with an infinitely thin flame, $\overline{\Sigma}$ is the mean flame perimeter per unit area. Bell et al. (2007) suggest the 2D method under-predicts $\overline{\Sigma}$ by 25% to 33%.

As can be seen in Fig. 4, the $\phi = 1.35$ flame is considerably taller than the $\phi = 0.7$ flame despite both possessing equal laminar flame speeds. This indicates that the diffusively stable nature of the rich flame significantly reduces the wrinkling and hence the turbulent burning velocity. The diffusively unstable ($\phi = 0.6$) flame is shorter than the stable one despite having a significantly lower laminar flame speed. This further confirms that the details of the correlation between local burning rates and stretch are important in determining the global flame properties.

Velocity statistics are conditioned on the instantaneous reactants (subscript r) and reported for $\overline{C} < 0.9$. The average z velocity was found to be zero, indicating the flow was 2D in the mean. The turbulence became isotropic a short distance downstream of the exit plane. As the flow proceeded into the flame brush, the fluctuations of the downstream component (u') increased due to local acceleration from gas expansion.

Profiles of the flame surface density were taken in the transverse (y) direction at the heights (x) indicated in Fig. 5. Near the base of the flame, the brush is thin for all cases with a full width at half maximum ranging from 1 to 1.1 mm. At downstream locations, the flame brush spreads and the flame surface is distributed over a greater volume.

The profiles of u' and Λ shown in Fig. 6 were measured along the centerline. The mean values were spatially averaged over five interrogation boxes in the y-direction. The longitudinal integral scale in the x-direction for each case is presented. These were computed only in the region near the burner exit. At locations well upstream the brush, the integral scales increased linearly with downstream distance. However, as the brush was approached the integral scale growth was affected by the flame surface as

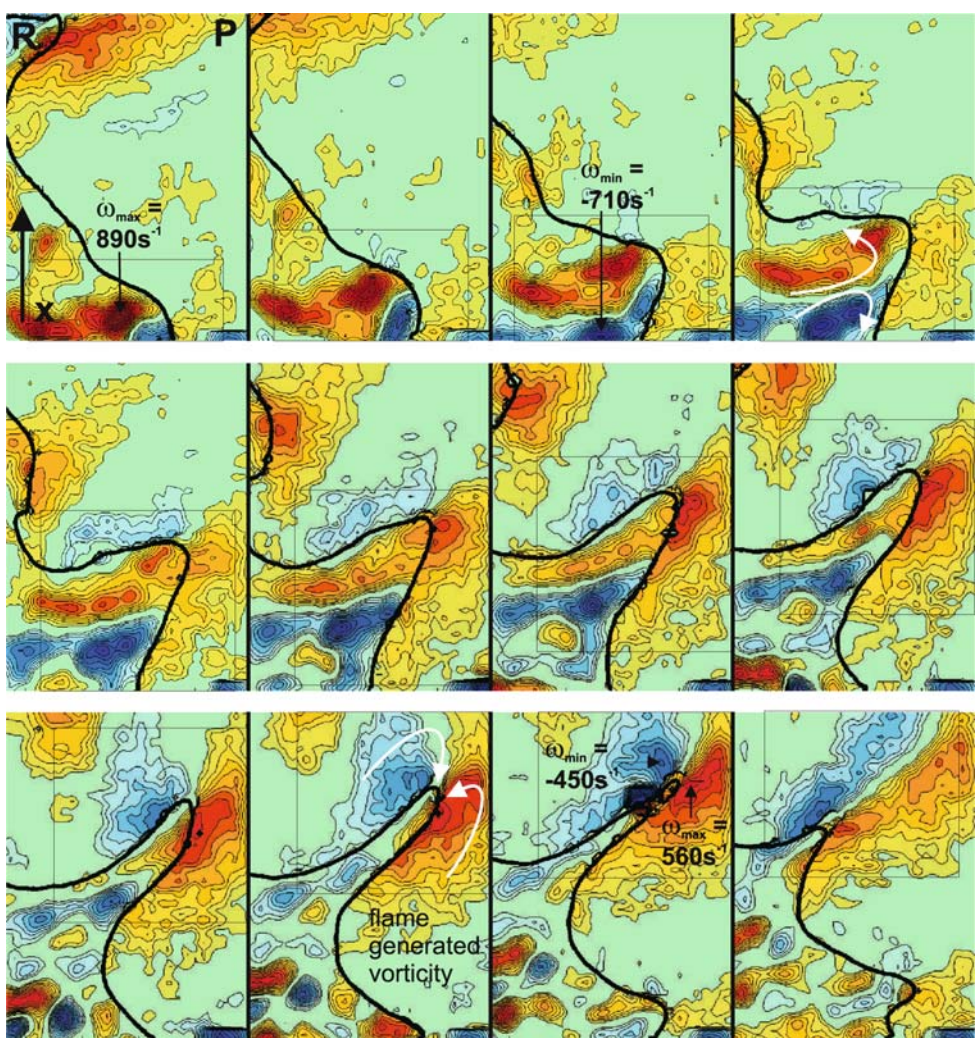


Fig. 7 Case 2-$\phi = 0.70$. Contours of vorticity between $-700\ s^{-1}$ (*blue*) and $700\ s^{-1}$ (*red*) with flame surface (*thick black line*). Field of view is 8.3 mm × 11.2 mm. Time between frames is 0.9 ms. A large wrinkle is formed by a pair of outwardly rotating eddies. Significant flame-generated vorticity is formed downstream of the wrinkle

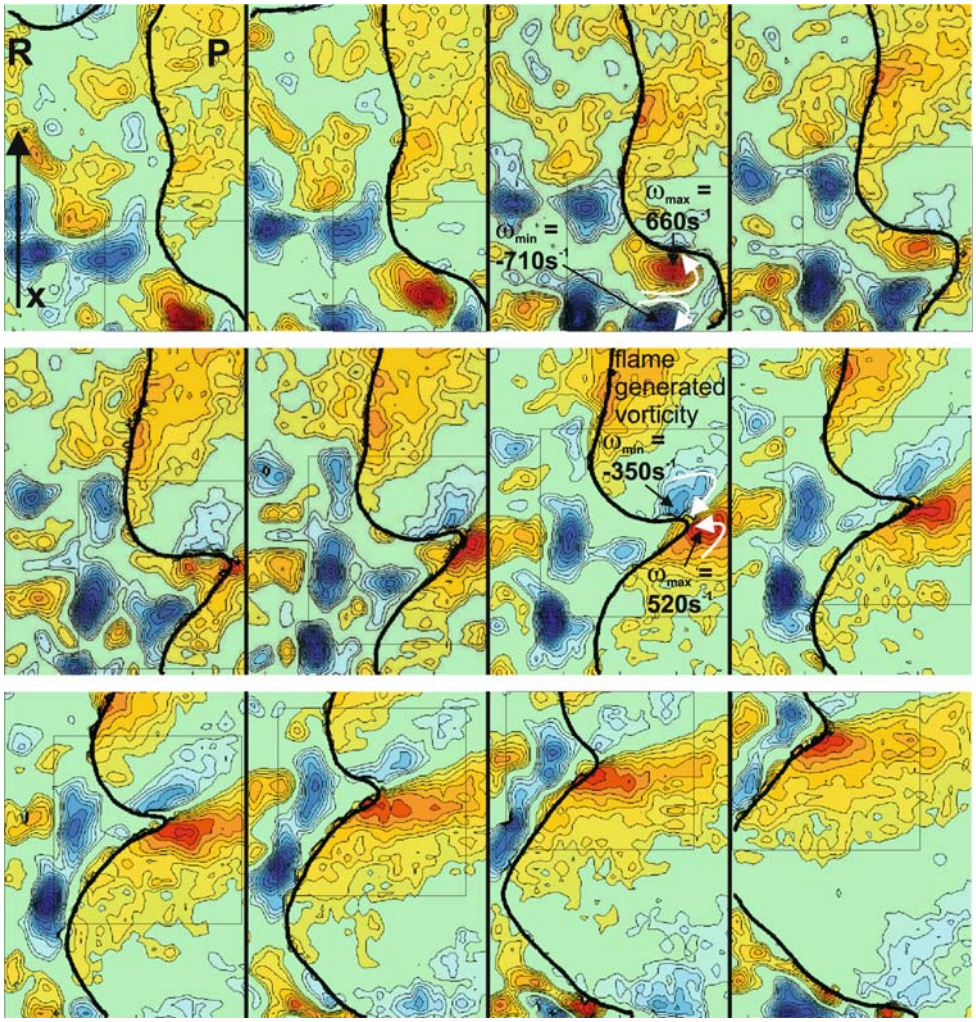


Fig. 8 Case 2-$\phi = 0.70$. Contours of vorticity between -700 s^{-1} (*blue*) and 700 s^{-1} (*red*) with flame surface (*thick black line*). Field of view is 8.3 mm × 11.2 mm. Time between frames is 0.9 ms. A wrinkle is formed by a pair of outwardly rotating eddies with similar strength to those in Fig. 10. Significant flame-generated vorticity is formed downstream of the wrinkle

shown for the short, $\phi = 0.7$ flame. This indicates that the flame surface can have significant effects on the velocity field a considerable distance upstream of the reaction zone as predicted by the theory of hydrodynamic instability (Williams 1985).

5.2 Flame dynamics

Presented in Figs. 7, 8, 9, 10, and 11 are time-histories of flame wrinkling phenomena observed using the CS-PIV technique developed. The events selected have low out-of-plane velocity. This ensures that both the eddies and flame wrinkles remain in the measurement plane during their interaction. Contours of vorticity in the z-direction (ω_z) between -700 and 700 s^{-1} are shown. The vorticity at any point was calculated by performing a line integral of the circulation around that point (Raffel et al. 1998). Red indicates positive vorticity (counter clockwise rotation) and blue indicates negative vorticity (clockwise rotation). The flame front is indicated by the thick black line. The reactants are to the left of the flame and the principal flow direction is upwards.

Figures 7 and 8 show typical wrinkling events for Case 2. An outwardly rotating eddy pair in the reactants impinges on the flame front causing a wrinkle to form around it. As the eddies pass through the flame, they are attenuated and large regions of FGV are created in the products. The FGV had the opposite sign of the incident eddies in the reactants. The wrinkle then forms into a cusp and burns through. In Fig. 7 the wrinkle propagates out of the field of view while in Fig. 8, a stable wrinkle shape is formed and propagates downstream.

This stable wrinkle shape is indicative of the importance of the hydrodynamic instability in turbulent flames. This phenomenon can be seen clearly in Fig. 9 for Case 1. Here, a large wrinkle is propagating in the absence of adjacent eddies. Velocity streaklines are shown relative to the mean downstream velocity in the reactants. These are computed by tracing the path of a fluid element that follows the

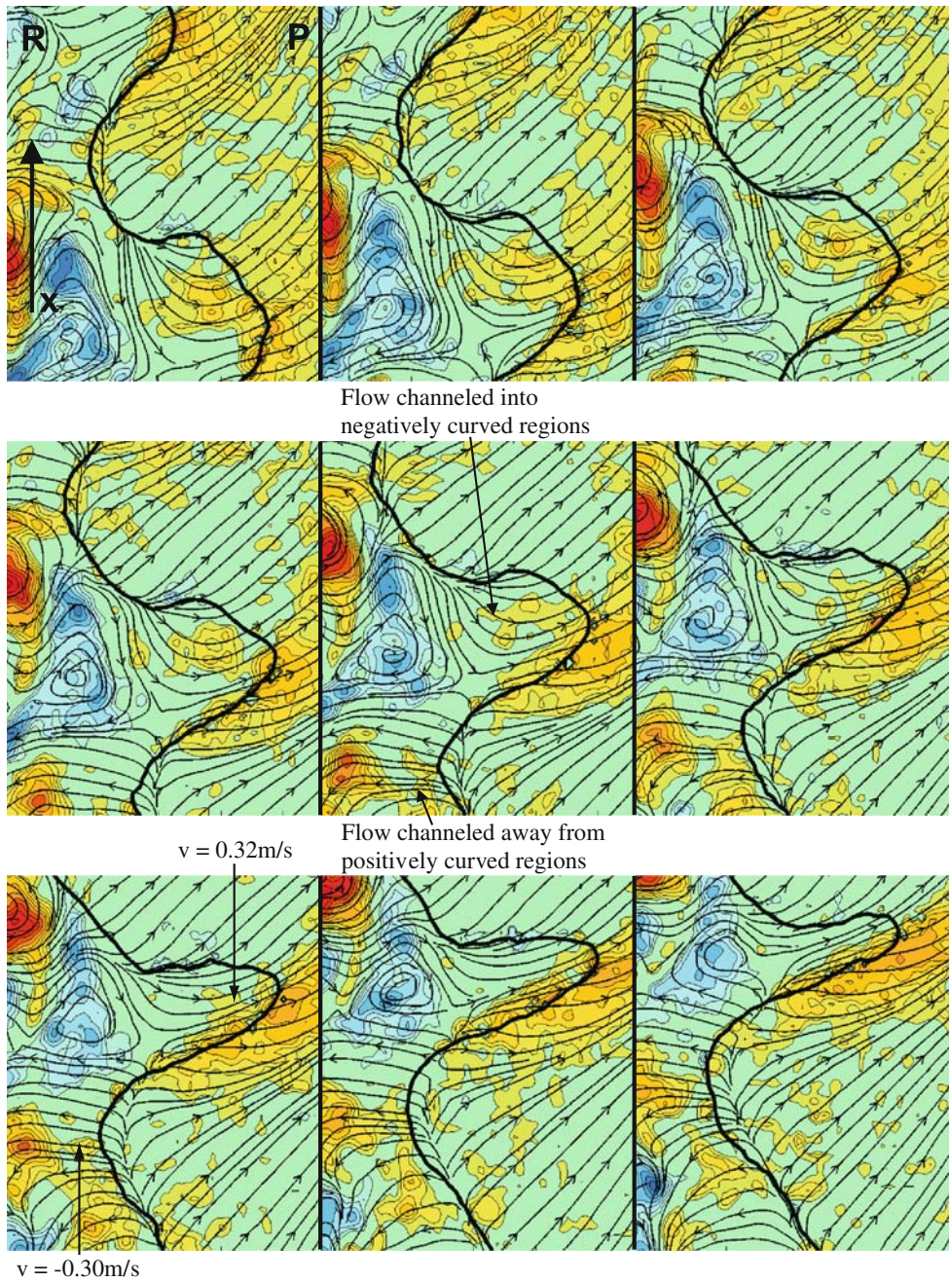


Fig. 9 Case 1-$\phi = 0.60$. Contours of vorticity between $-700\ s^{-1}$ (*blue*) and $700\ s^{-1}$ (*red*) with flame surface (*thick black line*). Field of view is 11.1 mm × 13.2 mm. Time between frames is 0.9 ms. Instantaneous velocity streaklines are shown, illustrating the mechanism behind the hydrodynamic instability

instantaneous velocity field represented in a single PIV image. As can be seen, flow is channeled into the the negative curvature region (concave towards the reactants) and away from the positive curvature regions (convex towards the reactants). The change in horizontal velocity between these regions was 0.62 m/s (frame 7 of Fig. 9), which is considerably larger than the laminar flame speed (0.15 m/s). This causes the flame winkle to elongate and narrow, with the positive sections being pulled into the reactants and the negative sections being pushed into the products. This is the behavior predicted by theory for 2D laminar flames (Williams 1985) and computations for 2D turbulent flames (Yuan et al. 2006).

Hence, for any wrinkle there will be competing effects that tend to cause it to either grow or attenuate. In addition, the propagation speed of the flamelet varies along the wrinkle. Differential diffusion between the deficient and excess reactants or the deficient reactant and heat can cause significant variations in the local flame speed (S_L) when a stretch rate is applied to a flame. If the correlation between stretch rate and flame speed is such that S_L is reduced in regions of negative curvature and increased in regions of

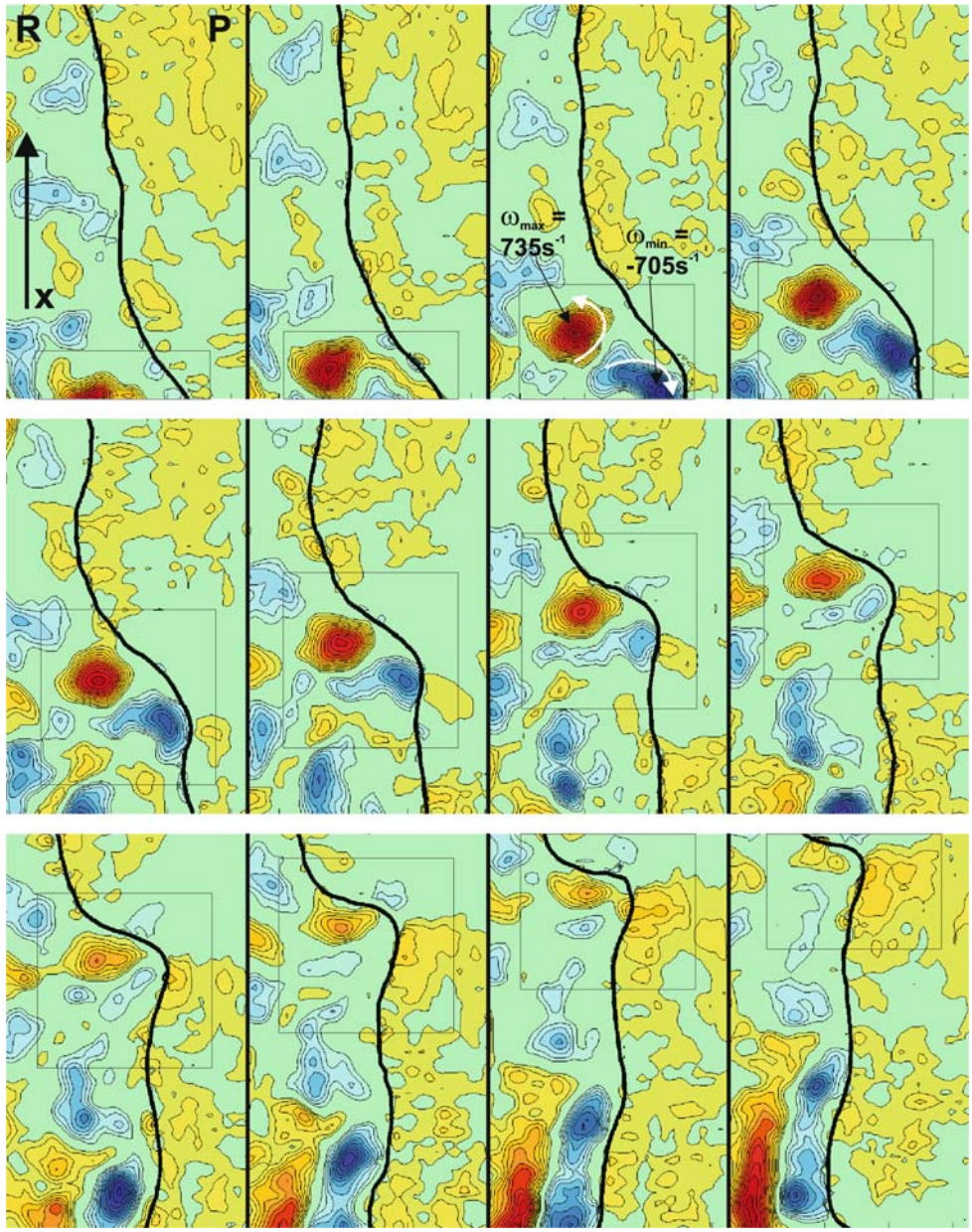


Fig. 10 Case 3-$\phi = 1.35$. Contours of vorticity between -700 s^{-1} (*blue*) and 700 s^{-1} (*red*) with flame surface (*thick black line*). Field of view is 8.3 mm × 13.6 mm. Time between frames is 0.9 ms. A wrinkle is formed by a pair of outwardly rotating eddies with similar strength to those in Fig. 8. The peak curvature is less than that of Fig. 8 due to preferential diffusion effects

positive curvature, the flame is easier to wrinkle and wrinkles tend to grow. If the opposite is true, the flame resists wrinkling. The importance of this mechanism can be seen by comparing Figs. 8 (Case 2) and 10 (Case 3). These cases have equal unstretched laminar flame speeds. In the absence of preferential diffusion effects on the flame speed, similar eddy structures should cause similar wrinkles. However, this is not the case. The flame in Fig. 10 is diffusively stable and resists the creation of high curvature, small scale wrinkles. However, high curvature wrinkles are created in Fig. 8, for which the conditions are diffusively neutral. Case 1 is expected to be slightly diffusively unstable and has a considerably lower laminar flame speed than Cases 2 and 3. Hence, the tendency for wrinkles in this flame to grow or propagate with large amplitude (as in Fig. 9) is a combination of these attributes.

The wrinkling events shown in Figs. 7, 8, 9 and 10 are typical for outwardly rotating eddy pairs. This configuration has velocity vectors that are diverging at the flame front and is the standard configuration studied in experiments and computations (Meneveau and Poinsot 1991; Colin et al. 2000; Mueller et al. 1998; Roberts and Driscoll 1991; Sinibaldi et al. 1998). However, inwardly rotating eddy pairs are very common in real turbulence and cause significantly different flame behavior. Figure 11 shows a wrinkle created by inwardly rotating eddies. In this case, the wrinkle has positive curvature (convex towards the reactants). The flame propagates outward relative to its center of

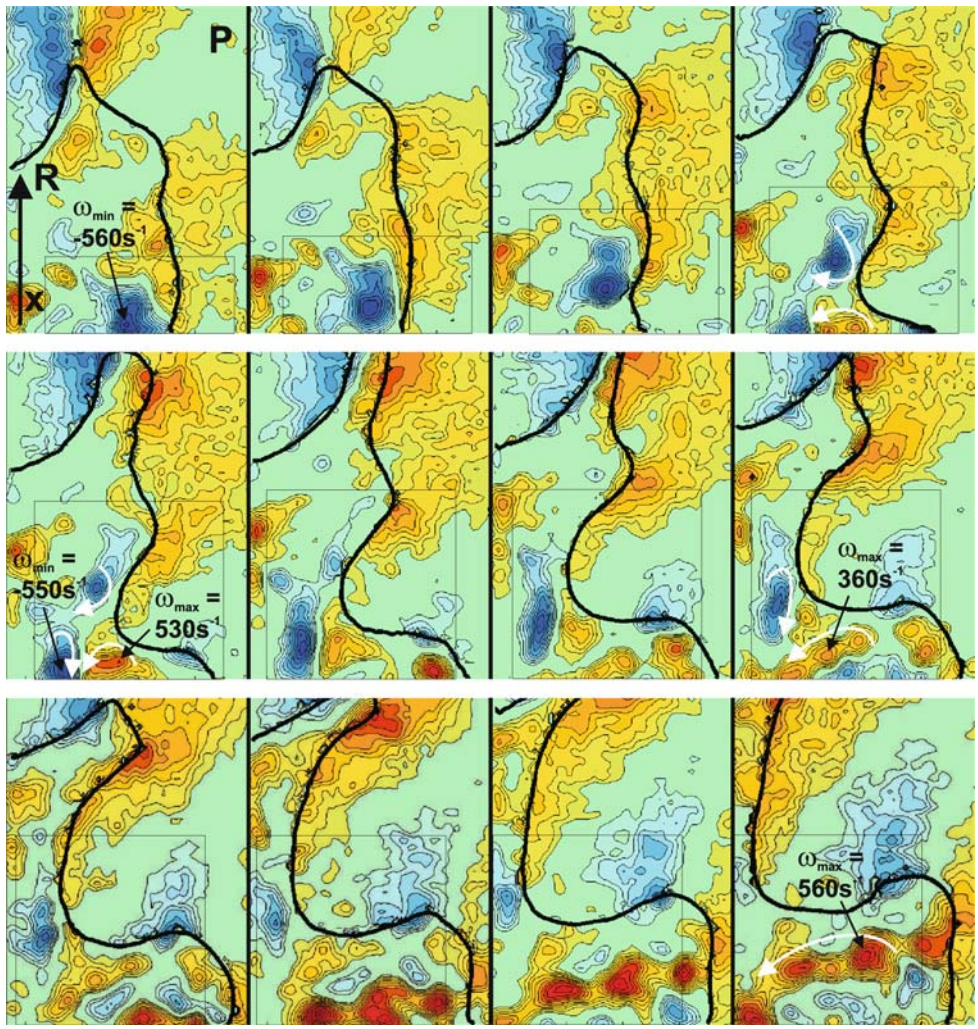


Fig. 11 Case 2-$\phi = 0.70$. Contours of vorticity between $-700\ s^{-1}$ (*blue*) and $700\ s^{-1}$ (*red*) with flame surface (*thick black line*). Field of view is 8.3 mm × 11.2 mm. Time between frames is 0.9 ms. A wrinkle having positive curvature is formed by inwardly rotating eddies. The wrinkle grows and encounters other eddies that affect its shape

curvature, causing the wrinkle to grow. As it grows, it encounters other eddies that effect the wrinkle shape.

Figure 12 shows a wrinkle formed around a group of four eddies. In isolation, each eddy pair would form a wrinkle with wavelength and amplitude corresponding to that particular pair. The upstream group, having relatively low peak vorticity, may have caused negligible wrinkling. However, the combination of eddy pairs form a large wrinkle with a wavelength corresponding to the outer dimensions of the entire group. The events shown in Figs. 11 and 12 illustrate how large groups of eddies interact in a different manner than individual pairs and clearly indicate the importance of studying eddy–flame interactions in a fully turbulent flow field with a dynamically stretched flame.

As mentioned above, there is considerable FGV in the products downstream of the highly negatively curved flame wrinkles. This FGV has been observed in both 2D laminar experimental studies (Mueller et al. 1998) and DNS (Louch and Bray 2001; Pan et al. 2002). The fundamental mechanism causing FGV can be deduced from the vorticity transport equation:

$$\frac{D\underline{\omega}}{Dt} = (\underline{\omega}\cdot\nabla)\underline{u} + \nu\nabla^2\underline{\omega} + \frac{1}{\rho^2}(\nabla\rho\times\nabla P) - \underline{\omega}(\nabla\cdot\underline{u}) + \underline{\xi} \tag{4}$$

where $\underline{\xi}$ represents the effects of viscosity gradients. Mueller et al. (1998) show that the source of FGV is the baroclinic torque, which is represented by the third term on the right hand side and caused by the misalignment of the density and pressure gradients across the flame. The density gradient is normal to the flame, requiring a tangential pressure gradient to create the torque. This tangential pressure gradient is due to hydrodynamic, hydrostatic, and flame curvature effects.

As can be seen from Figs. 7 and 8, there can be significant difference between the magnitudes of the positive and negative FGV, with the positive peak being up to 50% larger than the negative peak. This difference can be explained by the hydrostatic pressure gradient. Pan et al. (2002) have shown that the pressure gradient caused by flame curvature and reactant side velocity gradients points away from the peak negative curvature. Hence, for a flame

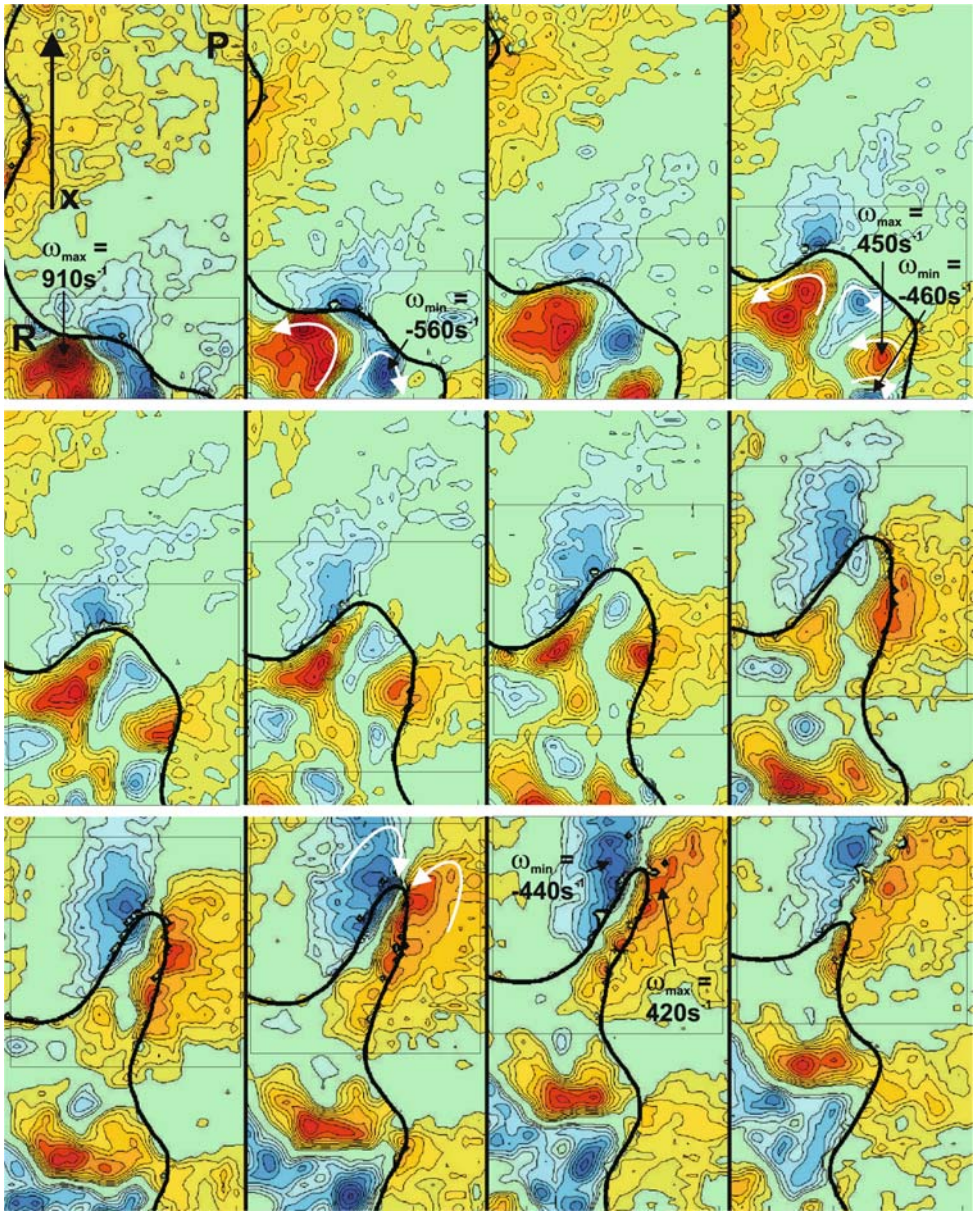


Fig. 12 Case 2-$\phi = 0.70$. Contours of vorticity between $-700\ s^{-1}$ (*blue*) and $700\ s^{-1}$ (*red*) with flame surface (*thick black line*). Field of view is 8.3 mm × 13.7 mm. Time between frames is 0.9 ms. A wrinkle is formed around a group of four eddies

wrinkle oriented vertically, the hydrostatic pressure gradient due to gravity acts to decrease the tangential pressure gradient on the upper portion of the wrinkle and increase it on the lower portion. This changes the baroclinic torque and produces the unequal FGV shown. This is further confirmed by comparing the positive to negative FGV disparities in Figs. 7, 8, and 12. As the wrinkle orientation rotates from vertical to horizontal, the difference between the negative and positive peaks disappears.

6 Conclusions

The unsteady dynamics of turbulent premixed flame wrinkling was observed for the first time using cinema stereoscopic PIV. This system is capable of obtaining three component vector fields at rates greater than a kilohertz with spatial resolution of 140 μm and an interrogation box size of 280 μm. Various wrinkling phenomena were recorded in turbulent slot Bunsen flames burning methane–air mixtures at various equivalence ratios. The flame surface was accurately determined from the gradient of the PIV particle image density. This contour of maximum particle density gradient was found to correspond to the maximum in gas density gradient.

Flame wrinkling caused by outwardly rotating eddy pairs was found to occur. A wrinkle was formed around the eddy pair, which was attenuated by the flame. Flame-generated vorticity having an opposite sign to that of the original eddies was created downstream of the negatively

curved wrinkle tips. The magnitude of the flame-generated vorticity was found to be dependant on the wrinkle orientation due to the hydrostatic pressure gradient.

The hydrodynamic instability was observed. Flow streaklines indicated that flow was channeled into the negatively curved flame regions and away from those positively curved. This is in agreement with 2D laminar theory. The velocities thus induced were found to be larger than the laminar burning velocity and therefore significant in the development of flame wrinkles. Further work is needed to quantify this effect. Preferential diffusion effects on the flame stability were also observed. The diffusively stable case exhibited marked resistance to high curvature wrinkling. This resulted in less flame surface area and a taller flame than the equivalent diffusively neutral case.

Wrinkling events involving large numbers of eddies were also observed. These occurred both as a series of inwardly rotating eddy groups causing a large positive curvature wrinkle and a foursome of outwardly rotating eddies causing a large negative curvature wrinkle. The significance of these events in generating flame surface area indicates that the canonical studies performed using isolated vortices may be insufficient to describe turbulent flame dynamics.

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